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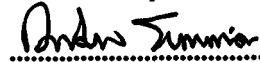
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METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH REED FDM SYSTEMS ON THE A.C. & D.C RAILWAYS

ENDORSEMENT & AUTHORISATION

Endorsed by:



A Simmons, Professional Head of Signal Engineering

Accepted for issue by:



P McAleer, Acting Head of Business Standards

Authorised by:



A McNaughton, Chief Engineer

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IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

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I. Introduction

I.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with reed FDM systems installed on the a.c. and d.c. electrified railway on Railtrack controlled infrastructure. This is achieved by:

- Providing a methodology to calculate the longitudinal voltage produced in the cable cores forming a reed FDM circuit.
- Providing a model to determine the voltage across a reed FDM receiver.
- Providing a methodology to define the permissible current in the traction supply.
- For reasonable worse case infrastructure conditions train current limits are defined.

This characterisation will take the form of a permissible train current which can then be compared to EMI the footprint of a piece of traction or rolling stock to enable a case to be made. Methods are included to analyse intact and faulty infrastructure.

The aim of this document is to provide a methodology for the demonstration of electromagnetic compatibility of reed FDM systems installed on Railtrack routes.

The interference mechanisms to be considered are conductive, capacitive and inductive coupling. The coupling is between the traction supply & return conductors and the lineside cables.

I.2 Scope

This document covers RR type reed FDM circuits installed on a.c. and d.c. electrified Railtrack controlled infrastructure.

Westinghouse FDM-NV and older type 'R' systems are not covered by this document.

I.3 Installation Types Covered by this Document

This document covers reed FDM systems installed in accordance with the reed FDM system manuals [refs. 6 & 7].

1.4 Equipment Types Included in this Document

The equipment covered within this document is the type 'RR' reed FDM equipment manufactured by GEC (now ALSTOM).

2. SOURCE DOCUMENTS

2.1 Group Standards

There are no Railtrack Group Standards relating to reed FDM systems. It is however assumed that all signalling equipment has been correctly installed and maintained in accordance with Railtrack group standards and procedures document GK/RT0231.

- 1 GK/RT0231 Signalling Maintenance Testing, issue 3, dated December 1999.
- 2 RTE/C/27010 Compatibility between electric trains and electrification systems, issue 1 November 1997.

2.2 Previous Safety Case Documentation

- 2 "Fleet Operation on the 25 kV AC Railway between Crewe and Doncaster – Reed FDM", AEA Technology Rail ref. I20-dr0458, version b0, dated November 1999.
- 3 "Class 373/1 Safety Assessment of the Reed FDM System on Class 373/1 Routes", AEA Technology Rail ref. I20-ds0016, version b0, dated April 1995.
- 4 "Class 373/1 Safety Assessment of the Reed FDM System on Class 373/1 Routes Supplement A to Appendix E - Probability of Occurrence of Failure Mechanisms for Interference Coupling", AEA Technology Rail ref. I20-dr0029, version a0, dated April 1995.

2.3 Other Documents

- 6 "Type RR Reed Frequency Multiplex System Handbook IBI38", GEC General Signal Ltd, dated October 1980.

"Type RR Reed Frequency Division Multiplex System Manual IBI65", GEC General Signal Ltd, dated November 1987.

3. Infrastructure Assumptions

The main assumptions that may be made while carrying out an analysis are summarised below.

- The infrastructure is assumed to be in the worst case for inductive coupling, i.e. the traction current flows in the conductor nearest the cable and returns in the conductor farthest from the cable. This is effectively a failed infrastructure condition for the length of the exposure, a pessimistic scenario.
- The infrastructure assumed is representative of the two track case.
- The FDM cable is assumed to be 1m away from the nearest running rail for the majority of systems. The case of a portion of the exposed cable being 0.3m away from the nearest running rail is also considered.
- The permissible transverse voltage is assumed to be 20% of the reed receiver drop away value [ref. 2].
- Minimum acceptable core to earth resistance for a cable is 500 k Ω [ref. 1].
- The cable forming the FDM circuit has a hard earth fault at one end and the minimum acceptable core to earth resistance at the other. Earth faults on Reed FDM systems are relatively rare. FRAME data indicates there were four earth faults on FDM cables in a five years [ref. 3]. This, coupled with a test periodicity of three months [ref. 1], mean that the scenario is highly pessimistic.
- Reed FDM circuits are carried in cores taken from the same layer of multicore cables [ref. 6,7].

4. Description of Equipment

4.1 Overview of Reed FDM Systems

A frequency division multiplex (FDM) system is used to carry a number of signals over a single pair of cable cores. These cores are commonly contained within multicore or

twisted pair cables. It is possible to carry up to 63 circuits in one pair of conductors. For each channel, an independent transmitter is provided which generates an a.c. signal at the particular frequency allocated to that channel; an individual receiver detects the frequency for each channel. The frequencies used for transmission are in the range from 350-1000 Hz. The discrete channels within the range have been chosen to avoid supply frequency harmonics. The channel numbers of the Type RR system are grouped as follows:

- f1 – f51: 51 channels in the range of 395.25Hz to 879.50Hz, originally used for vital systems but not normally used as such adjacent to electrified railways. The channels are however used for non-vital systems alongside electrified lines;
- f81 – f92: 12 ‘non-vital’ channels in the range of 383.25Hz to 842Hz.;
- f71 – f72: 2 ‘spare’ channels;
- f401 – f416: another set of 16 ‘vital’ channels for 50Hz railways, in the range 410.25Hz to 817Hz, with frequencies designed to avoid both even and odd harmonics of the mains.

The frequencies used for transmission are thus broadly in the range from 383 - 880 Hz.

The transmitter and receiver in the reed FDM system consist of a reed filter and an amplifier. The reed filter is a band-pass filter, which passes only the frequency allocated to the particular channel and excludes all others. The selectivity of the reed filters is such that the system can be operated with frequencies as close as 4 Hz; hence a very large number of channels can be accommodated within a small range of frequencies.

A full technical description of the operating characteristics of reed FDM systems is given in GEC manuals IB138 and IB165 [refs. 6 & 7].

4.2 The Reed Filter

The reed filter contains a pair of tuned reeds. Each is a short bar of a special alloy, clamped at one end and shaped so as to vibrate at a specific frequency, which is set during manufacture. It is placed such that it lies along the axis of a coil, with its free end between the poles of a permanent magnet, as illustrated in figure 1.

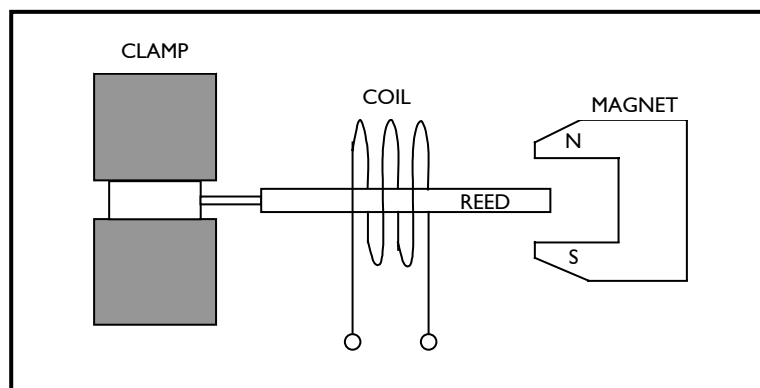


Figure 1: Basic Reed

If a current flows in the coil, it magnetises the reed, which is therefore displaced by the field of the permanent magnet. If the current is made to vary sinusoidally the reed will oscillate in the field. At most frequencies the amplitude of the oscillation is small, but if the frequency of current flowing in the coil coincides with the natural frequency of vibration of the reed, the amplitude becomes very large. Alternatively, if the assembly is vibrated mechanically, the reed will oscillate between the poles of the magnet, causing a voltage to appear between the terminals of the coil. This voltage is small unless the frequency of the impressed vibration is the same as the resonant frequency of the reed, in which case it will become much larger.

The complete reed filter consists of two of these reed assemblies tuned to the same frequency, fastened together. An a.c. current in one coil will cause the reed of the other coil to oscillate, hence inducing a voltage in the secondary coil. This voltage is only significant if the reeds are oscillating at their resonant frequency. The device can therefore be seen to act as a band-pass filter; only signals having the frequency to which the reeds have been tuned are coupled through the coils.

4.3 System Classification

The reed FDM systems can be implemented as vital or non-vital systems depending on the application.

4.3.1 Vital Reed FDM System

Vital reed FDM systems are used for the control and indication of safety critical functions between an interlocking and trackside signalling equipment. They are normally used in fairly limited applications in specific areas. Vital systems generally utilise twisted pair cabling to transmit the signals between the trackside equipment and the interlocking, although multicore is sometimes employed. When multicore cabling is used the two cores that form the circuit must be taken from the same layer within the cable.

The trackside equipment may be some distance from the interlocking; the use of vital FDM provides a method by which the interlocking can control signalling equipment at a greater distance more economically than if individual circuits were required by each function. A typical layout of vital reed FDM systems is shown in figure 2 below.

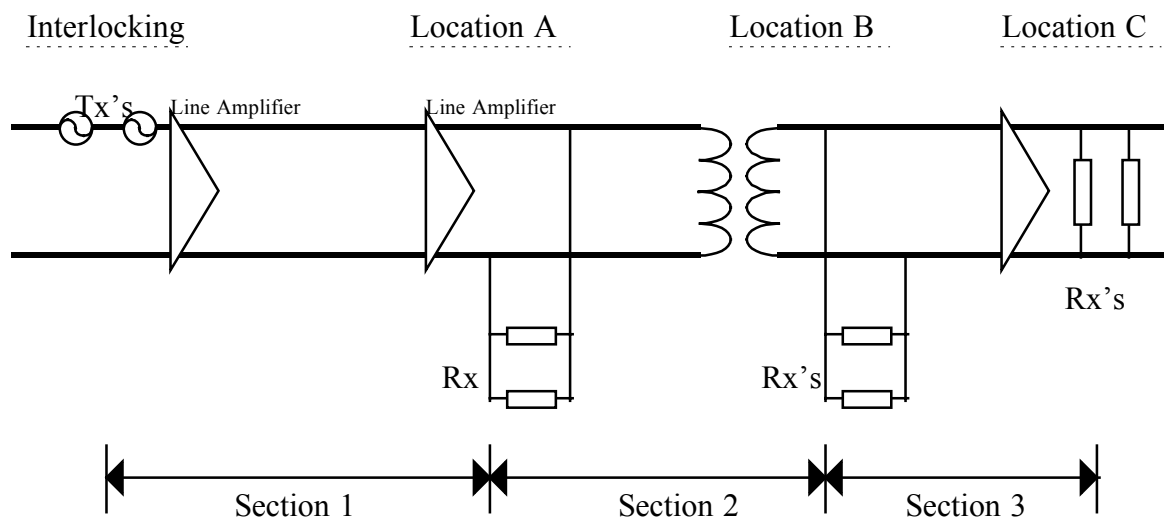


Figure 2: Typical vital reed FDM system

The system shown in figure 2 is typical of a control system with transmitters at the interlocking. An indication system would have the transmitters distributed throughout the system with receivers concentrated at the interlocking.

The line is sectionalised to reduce the effect of interference induced into the cable cores. The maximum permitted length of a section is 2 km. The line is sectionalised by the use of line isolating transformers and line amplifiers. The line amplifier is a uni-directional amplifier that is used to counter-act the attenuation in the line. The voltage gain of the amplifier is not greater than 2 [ref. 6].

This sectionalisation means that transverse voltages present in one section of the system should not be additive with the following section because a phase shift, nominally a reversal, will occur at isolation transformers. In very long systems the slightly non-perfect cancellation at sectionalisation points, that is not significant in short systems, may become significant.

4.3.2 Non-Vital Reed FDM Systems

Non-vital reed FDM systems are essentially similar to the vital systems but may incorporate equipment which is not necessarily to 'fail - safe' standards. Multicore cable may be used rather than twisted pair for transmitting the signals. The non-vital systems are used for transmission of less safety critical controls and indications to and from the signalman's console.

5 INFRASTRUCTURE ASSUMPTIONS

The methodology presented utilises the following assumptions:

- The traction current flows in the conductor that is closest to the FDM cable. This will lead to the maximum possible longitudinal voltage due to this current flow being calculated.
- All of the traction current returns to source via the conductor that is furthest from the FDM cable. This leads to pessimistic results since this will result in the lowest longitudinal voltage being calculated for this current flow. The resultant longitudinal voltage is the difference between that calculated for the 'go' and 'return' current, therefore this assumption leads to the greatest difference being calculated.
- A section length of 4km is assumed, this allows for the non-perfect cancellation of common mode voltages at one sectionalisation point.
- The voltage induced into the cores of a twisted pair cable or two cores from the same layer of a multicore cable is the same [ref. 2].
- A factor of 2 is included to account for the presence of a line amplifier immediately preceding the receiver in question, as discussed in section 3.3.1.
- Carson Pollaczek equations are suitable for the calculation of longitudinal voltages in the railway environment [ref. 2].

6 Susceptibility Analysis

6.1 Hazards to Reed FDM Systems

Two effects of interference must be considered, namely:

- that which could cause a reed receiver to be energised in the absence of a

transmitted signal

- that which would cause a reed receiver to de-energise its output when the transmitted signal is present

False operation of the reed FDM systems may result from interference coupled into the FDM cabling, which produces a voltage across the receiver. Interference may be coupled into the cabling by inductive, capacitive or conductive coupling between the cable cores and the traction supply system.

6.2 Coupling Mechanisms

There are three coupling mechanisms that must be considered between the traction supply system and the reed FDM cabling: inductive, capacitive and conductive. Each of these mechanisms can result in a transverse voltage across the reed FDM receiver.

6.2.1 Conductive Coupling

For conductive interference to affect the reed FDM systems there would have to be two or more earth faults on the circuit. This would result in a considerable increase in crosstalk levels, and is fundamentally dangerous for vital circuits. It is likely to result in a failure of the FDM system and is therefore not considered further.

6.2.2 Capacitive Coupling

The calculation of capacitive coupling is carried out using conventional image theory, which is based upon considering the reflection image for each conductor into the earth. The worst case conditions used for calculating the capacitive coupling are fundamentally different to those used for inductive coupling. In the case of capacitive coupling, the calculations assume that there is an open circuit at the transmitter end of the reed FDM circuit whereas, in the case of inductive coupling, the calculations assume that there is a low impedance at the transmitter end of the circuit. Since the transmitter has a relatively low impedance, 20Ω from ref. 6, this coupling mechanism is considered to pose less of a risk to reed FDM systems than the inductive coupling mechanism considered below.

It has also been argued that the two infrastructure failures required for this coupling mechanism to pose a threat to Reed FDM systems have a remote probability of occurring simultaneously on the same Reed FDM circuit (ref. 4, 5). Further protection from this mechanism is provided by the requirement of an open circuit at the transmitter end of the circuit which would result in maloperation of the system.

6.2.3 Inductive Coupling

Inductive coupling has the potential to produce voltages in the cables forming the reed FDM system. Different voltages induced into the two cores connected across an FDM receiver may result in a transverse voltage across the receiver.

The inductive coupling between the traction supply systems and the reed FDM cable will result in longitudinal voltages being induced into the cores that form the circuit. The worst case condition that must be considered, other than the two earth faults discussed in section 5.2.1, is that when the insulation of the reed FDM cable is degraded or has a partial earth fault and a solid earth fault occurs on the other end of the cable. The circuit representation of this situation is shown in figure 3.

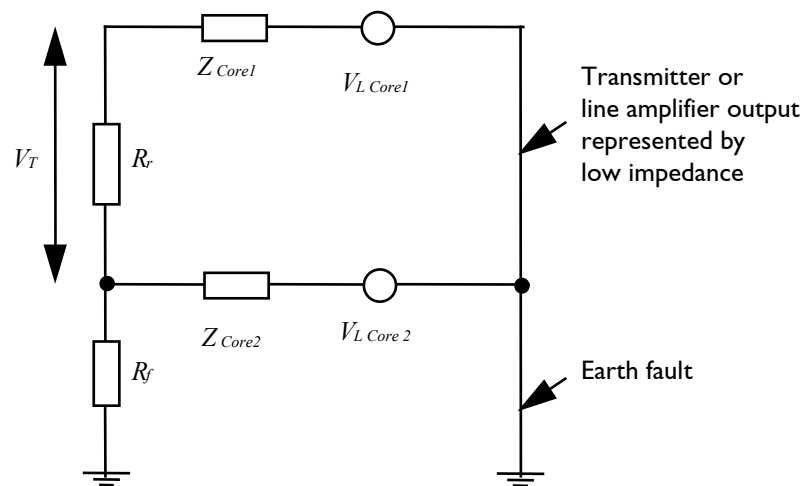


Figure 3: Simplified electrical circuit representation of reed FDM system

- Where:
- V_{LCore1} Is the induced longitudinal voltage in core 1 of the reed FDM system
 - V_{LCore2} Is the induced longitudinal voltage in core 2 of the reed FDM system
 - Z_{Core1} Is the impedance of core 1 of the cable
 - Z_{Core2} Is the impedance of core 2 of the cable
 - V_T Is the induced transverse voltage across the reed FDM receiver
 - R_f Core to earth resistance
 - R_r Is the equivalent input resistance of the reed FDM receiver

The input resistance of the reed FDM receiver is $1.1\text{k}\Omega$ [ref. 7]. The minimum acceptable core to earth resistance is $500\text{k}\Omega$ [ref. 1], hence values lower than this need not be considered. The earth fault is assumed to be of zero resistance.

The simplified circuit shown in figure 3 does not take into account the effects of capacitive leakage from the cable cores to earth. At reed FDM frequencies the effect of this can be significant and hence needs to be included to determine train current limits.

From testing carried out on multicore and twisted pair cables carried out [ref. 3], it was found that the voltage induced into two cores from the same layer of a multicore cable or two cores forming a twisted pair is the same. This is due to the 'lay' or twist of the cable being short relative to the length of the section. Thus in the circuit shown in figure 3 the longitudinal induced voltages for both cores are the same and are independent of the construction of the cable used to form the reed FDM circuit. This is only the case for interference external to the cable, for interference between adjacent pairs (crosstalk) the behaviour of the two types of cables can be very different.

6.3 FDM System Susceptibility

6.3.1 Transverse Voltage Limits

The minimum threshold transverse voltages across the reed receiver to be applied are as follows:

- vital systems set at 20% of the receiver drop away voltage (112 mV ref. 6) giving 22.4 mV,
- non-vital systems set at 60% of the receiver drop away voltage (63 mV ref. 6) giving 37.8 mV.

These levels are taken to be those to cause a wrong side failure of the reed FDM system. The level that will be used to determine the susceptibility will be the lower limit of 22.4mV. This level is lower than that which would be required to cause a right side failure of the system, i.e. de-energise the receiver when it should be energised since this is lower than the difference between the pick up and drop away levels of the receiver. Thus the limits for wrong side failure used here envelop the limits for right side failure of the system.

6.3.2 Reed FDM System Model

To determine the permissible current at the rail/wheel interface of the train, the

transverse voltage across the receiver for 1 A of current flowing in the inducing circuit can be determined. This is done by substituting the longitudinal voltage per 1 A of interference current into the equivalent circuit shown in figure 4. Once this transverse voltage is calculated per 1 A of interference current it is simply a case of applying the factor of 2 for a line amplifier as discussed in section 4, and scaling the inducing current up or down to find that which would give rise to a transverse voltage of 22.4 mV across the receiver.

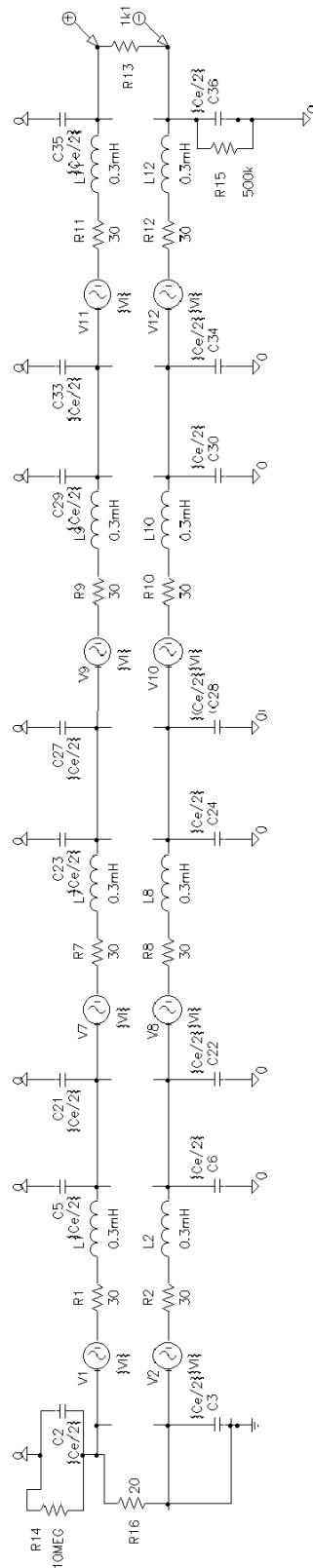


Figure 4: Circuit representation of Reed FDM circuit

From figure 4 it can be seen that the core pair has been sectioned into 4, each section representing 1km of the circuit. This allows for the treatment of distributed capacitance and induced voltage.

The capacitance to earth of each core is set to 4nF/km [ref. 3]. The receiver is modelled with a 1.1k Ω resistor [ref. 5] and the transmitter by a 20 Ω resistor [ref. 6]. The resistance of the earth fault at the receiver is taken as 10M Ω (effectively open circuit) or 500k Ω (the specified minimum acceptable cable to earth resistance, ref. 1). The resistance of the fault at the transmitter is assumed to be zero, a hard earth fault. The longitudinal voltages used are the longitudinal voltages per km per 1A of interference current. When these voltages are substituted into the model the transverse voltage per 1A of interference current can be calculated. From this the permissible interference can be calculated as described above.

6.3.3 System Topology (Normal Systems)

The topology of the traction supply system and the location of the reed FDM cable with respect to it will determine the levels of induced longitudinal voltage. The reed FDM cable is commonly run in troughing in the cess alongside the railway. For the maximum levels of induced voltage the current must 'go' on the conductor nearest to the cable and 'return' on the conductor furthest from the cable.

Using this, the worst case that is considered likely on the a.c. railway is the case when 'go' conductor is the nearest traction return rail and the return conductor is the furthest overhead line on two track railway. This is effectively a case of failed infrastructure; no traction return current is flowing in the track on which the train would be located. This topology is shown in figure 5.

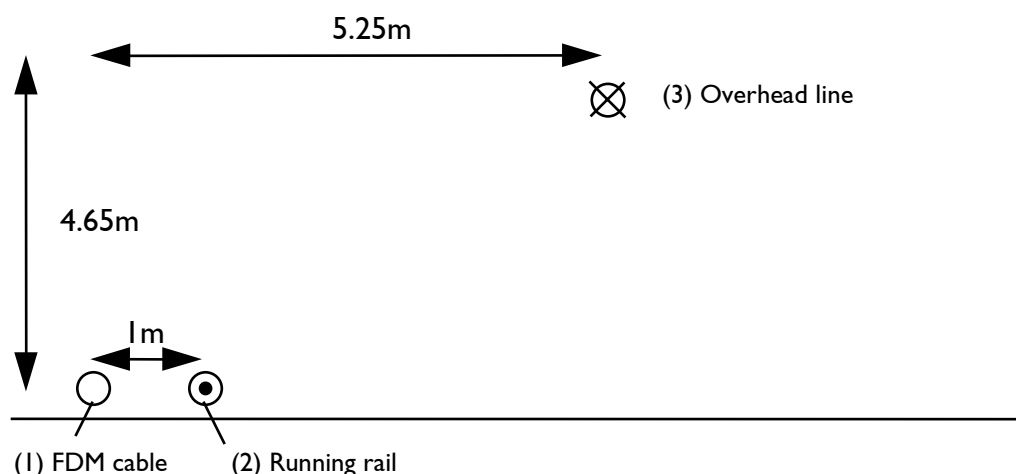


Figure 5: Topology of traction supply system and FDM cable modelled for the a.c. railway

On the d.c. railway a reasonably worse case topology is shown in figure 6 where the traction current flows in the running rail nearest to the reed FDM cable and the farthest conductor rail.

The slight height offset (76mm, ref. 2) of the conductor rail will have negligible effect on the calculated induced voltage.

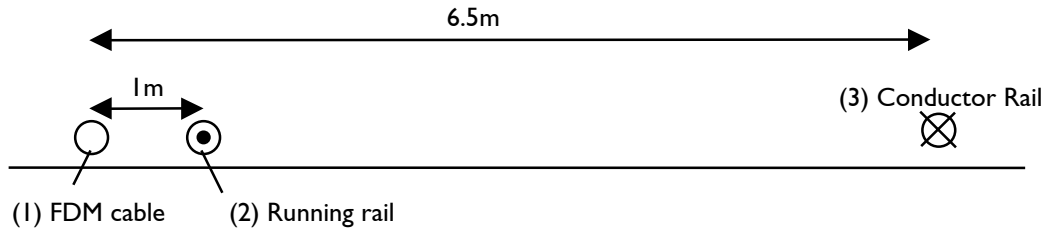


Figure 6: Topology of traction supply system and FDM cable modelled for the d.c. railway

6.3.4 Longitudinal Voltage Calculation

The method used to calculate the voltage induced into the cable cores forming the reed FDM circuit is the Carson Pollaczek equation, in the case of the inducing circuits described in section 5.3.3 this equation can be reduced to the following form:

Due to the nature of the inducing circuit, i.e. one 'go' and one 'return' conductor, the calculated induced longitudinal voltage in the cable core will not be affected by the soil resistivity used in the calculation.

$$V_l = j\mu_0 f I \left(\ln \left(\frac{D_e}{D_{21}} \right) - \ln \left(\frac{D_e}{D_{31}} \right) \right) \quad (\text{V/m})$$

where:

$$D_e = 659 \sqrt{\frac{\rho}{f}}$$

ρ	=	soil resistivity (Ωm)
I	=	Current in inducing circuit (A)
f	=	Frequency (Hz)
μ_0	=	$4\pi \times 10^{-7} \text{ H/m}$

D_{21} = distance between conductor 2 and conductor 1

D_{31} = distance between conductor 3 and conductor 1

Note the conductor assignments are those given in figures 5 and 6 where conductor 1 is the FDM cable core and conductors 2 and 3 form the inducing circuit.

It is hence apparent that the nature of the inducing circuit analysed means that the induced longitudinal voltages are independent of the soil resistivity (ρ) used in the calculation.

In this case studied here where $I = 1\text{A}$ the equation V_l simplifies to:

$$V_l = j\mu_0 f \left(\ln \left(\frac{D_{31}}{D_{21}} \right) \right) \quad (\text{V/m/A})$$

For substitution into the model shown in figure 4 the values from this equation are simply scaled up to give longitudinal voltages per km. For the topologies shown in figures 5 and 6 for the a.c. and d.c. railways respectively the longitudinal voltages over the range of reed FDM frequencies is as shown in table 1.

Frequency (Hz)	A.C. railway (V/km/A)	D.C. railway (V/km/A)
350	0.86	0.82
380	0.93	0.89
410.25	1.01	0.97
414.75	1.02	0.98
419.75	1.03	0.99
425.75	1.05	1.00
430.25	1.06	1.01
462	1.13	1.09
466.5	1.14	1.10

473	1.16	1.11
477.5	1.17	1.12
568.75	1.40	1.33
620.5	1.52	1.46
624.5	1.53	1.47
663	1.63	1.56
765.25	1.88	1.80
770	1.89	1.81
817	2.01	1.92
850	2.09	2.00
900	2.21	2.12
950	2.33	2.24

Table I: Calculated induced voltages for the a.c. and d.c. railway topologies

From table I it is evident that the induced voltages on the a.c. and d.c. railway are similar, this is due to the dominant effect of the current flowing in the nearest conductor. In both cases this is assumed to be 1m away from the cable leading to similar results.

6.3.5 Receiver Transverse Voltage Calculation

The receiver transverse voltage is calculated by substituting the longitudinal voltages calculated in section 5.3.4 into the model shown in figure 4.

The transverse voltage could be affected by the value used for the minimum core to earth resistance used in the calculation. Sample calculations are shown in table 2 for a 500k Ω to earth fault and 10 M Ω (effectively open circuit) for frequencies of 350Hz and 950Hz.

Earth fault resistance	Transverse voltage per 1A of interference current @ 350 Hz	Transverse voltage per 1A of interference current @ 950 Hz
500 k Ω	1.148mV	7.13 mV
10 M Ω	0.96mV	7.07 mV

Table 2: Transverse voltages at the receiver for 350Hz and 950Hz for minimum core to earth resistance and open circuit cases based on 4km exposure length

From the results presented in table 2 it is evident that at higher frequencies the earth fault resistance has less effect on the transverse voltage. This is due to the decreased reactance of capacitive leakage at these higher frequencies and demonstrates that a single hard earth fault on an FDM system can have a great effect on the system susceptibility when the effect of capacitive leakage is taken into consideration. Therefore the measures outlined in GK/RT0231 are essential to maintain the integrity of reed FDM systems and their susceptibility to external interference from all sources. The system can still function normally with a single earth fault.

If the calculation is carried out for the full range of reed FDM frequencies then the results obtained are as shown in table 3.

Frequency (Hz)	Transverse voltage at receiver per 1A of interference current for a.c. case (mV)	Transverse voltage at receiver per 1A of interference current for d.c. case (mV)
350	1.14	1.10
410.25	1.51	1.44
466.5	1.89	1.81
568.75	2.71	2.60
663	3.61	3.46
817	5.37	5.14
900	6.47	6.20
950	7.13	6.88

Table 3: Results for the transverse voltage at the receiver for the a.c. and d.c railway topologies

6.3.6 Line current Limit Calculation

The permissible interference current can be found by dividing the permissible transverse voltage by the transverse voltage per 1 A of interference current and then dividing this result by 2 to account for the presence of a line amplifier (see section 4).

$$\text{Permissible Interference Current} = \frac{22.4 \text{ (mV)}}{2 \times \text{Transverse Voltage (mV/A)}} \text{ (A rms)}$$

Applying this scaling to the transverse voltages in table 4 give the permissible interference currents shown in table 5.

Frequency (Hz)	Permissible interference current for the a.c. case (A rms)	Permissible interference current for the d.c. case (A rms)
350	9.84	10.17
410.25	7.43	7.76
466.5	5.93	6.18
568.75	4.14	4.31
663	3.10	3.24
817	2.09	2.18
900	1.73	1.81
950	1.57	1.63

Table 4: results for the transverse voltage at the receiver for the a.c. and d.c railway topologies

These results are also presented in figure 7.

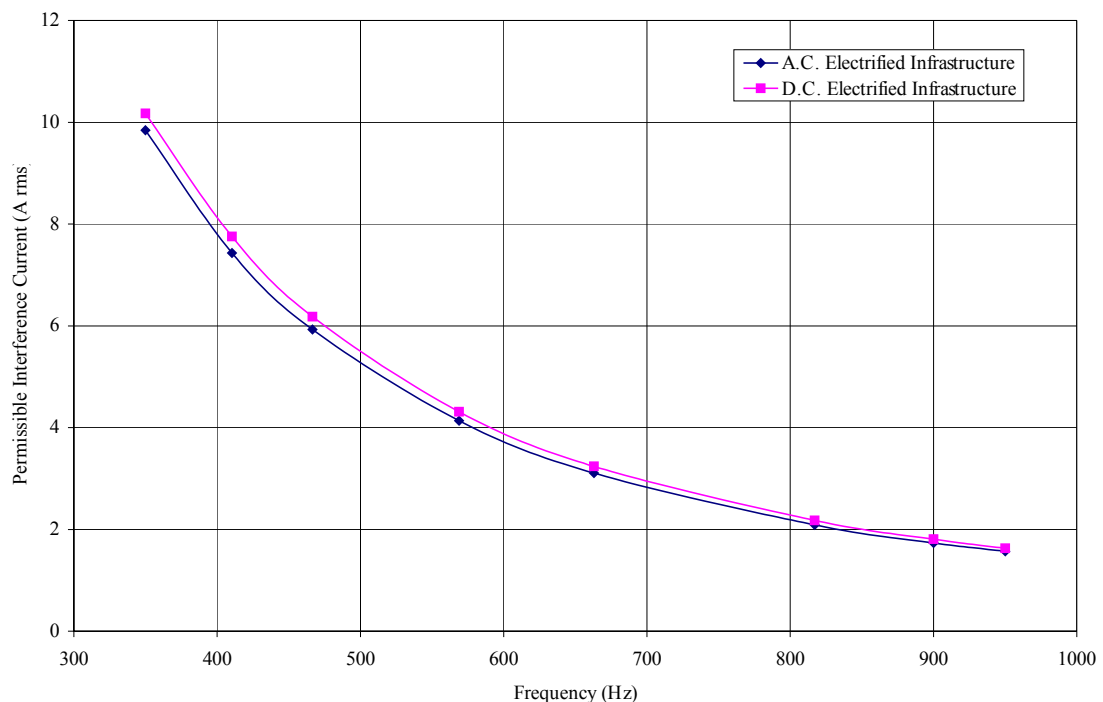


Figure 7: Permissible interference current for reed FDM systems adjacent to a.c. and d.c. electrified infrastructure

It should be noted that although the permissible current is presented here as a smooth curve, it is in fact made up of a combination of the response of the reed receivers in the range. As such it represents a line joining the minimum permissible current at all of the reed frequencies. The permissible current in between the discrete channels will be significantly higher than that shown on the graph, except where other systems impose lower limits (as for example reed track circuits in the range 357 to 429Hz).

6.4 Reed FDM cable laid close to railways with traction return/reinforcing conductors

There may be occasions where the cables within a Reed FDM system are located very close to traction return or reinforcing conductors. This may occur where a line has been electrified after the Reed FDM system has been installed or where there is limited space alongside the railway.

This situation will result in greater induced voltages in the cores of the cable forming the reed FDM system, the equation in section 6.3.4 predicts that this will be the case. In order to evaluate the effect that this would have on the analysis presented in section 6.3 the topologies shown in figures 8 and 9 have been used to account for this proximity over a 1km section of the total considered reed FDM circuit length of 4km.

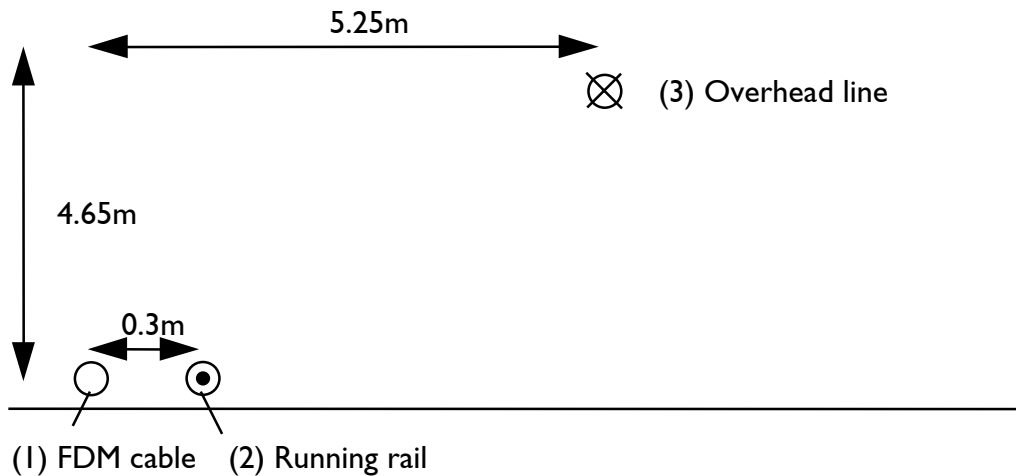


Figure 8: System topology for close proximity (a.c. railway)

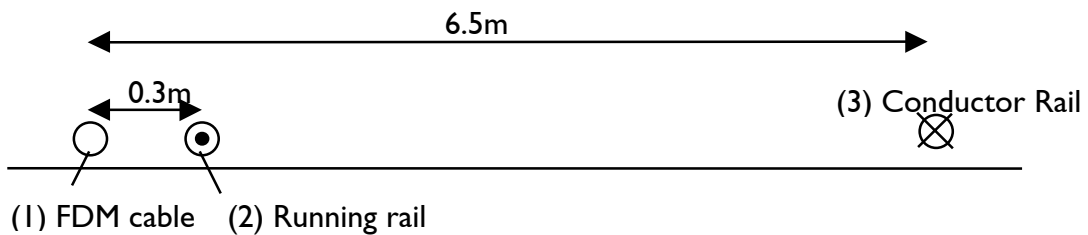


Figure 9: System topology for close proximity (d.c. railway)

The circuit shown in figure 4 can be used to analyse these scenarios by changing the voltage sources for one of the sections to the longitudinal voltage calculated using the topologies in figures 8 and 9 for the a.c. and d.c. railways respectively. Using the same compatibility criteria as in the previous case the curves of permissible line current shown in figure 10 can be calculated. These curves are based on the same permissible transverse voltage as the previous case and the same factor of 2 for a line amplifier.

The results shown in figure 10 are pessimistic for the case of a traction strengthening conductor since in this case not all of the current would flow in the conductor nearest to the FDM cable. This holds true for all cases where current can return to source via more than one conductor be these rails or cables.

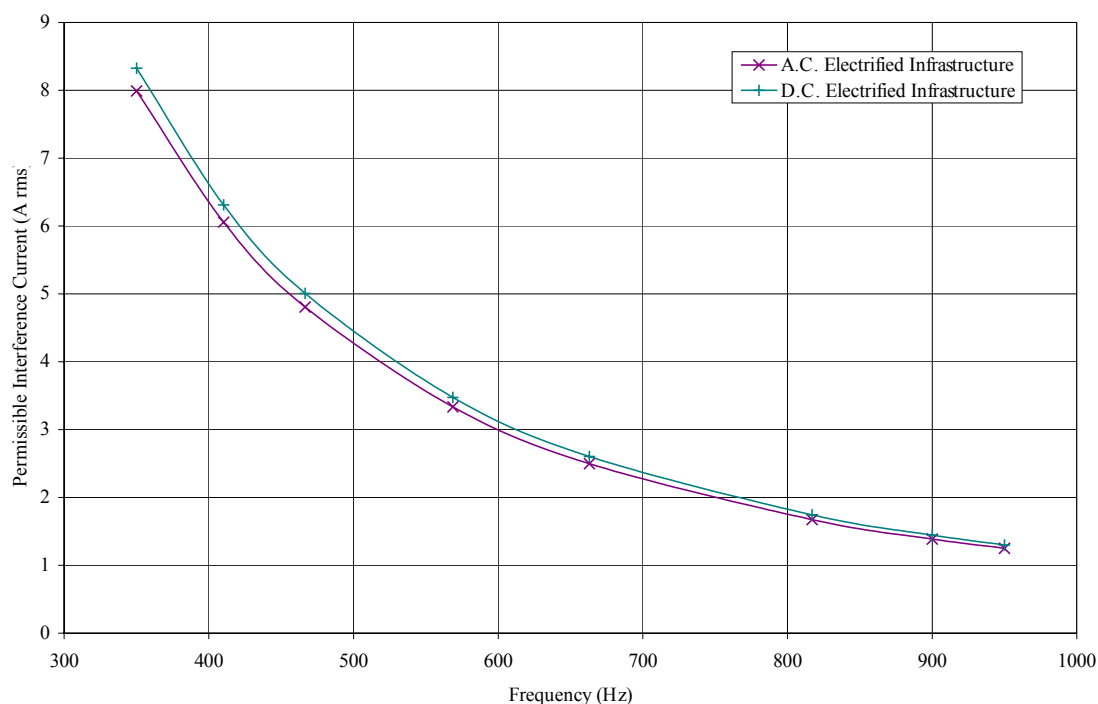


Figure 10: Permissible interference current for reed FDM systems with sections in closer proximity to a.c. and d.c. electrified infrastructure

Comparing figures 7 and 10 it can be seen that, as expected, moving the cable closer to the infrastructure results in lower permissible currents. The currents calculated here are based on the same pessimistic assumptions and budgeting applied for the previous case. Since these currents are some 15% lower than calculated in figure 7 a slight relaxation in budgeting and/or allowance in the factor of 2 for the presence of a line amplifier would result in the same limits as in figure 7. Such proximities are expected to be relatively rare. The Reed FDM circuit in question should be examined for long (>1km) sections in close proximity to the nearest traction current carrying infrastructure if these are found, limits should be recalculated and compared to the permissible currents given in section 7. Again it must be borne in mind that the system under analysis is effectively faulty.

6.5 Long Reed FDM Systems

As mentioned earlier where the Reed FDM circuit is very long then the non-perfect cancellation of potentially additive voltages at sectionalisation points may become significant. Due to the nature of the interference mechanism, in most cases the length of the electrical feeding section mitigates this. Typically feeder sections on the a.c. railway are in the order of 20km long meaning that the maximum exposure length is limited to this length. In the case of the d.c. railway substation spacing is typically of the order of 4-6km, in inner suburban areas this spacing may be much less. Where

very long (>40km) Reed FDM circuits exist the additive effects should be considered, attention should be paid to the interference generation mechanisms within the rolling stock and whether these are locked to supply frequency.

7 METHODOLOGY FOR DEMONSTRATION OF COMPLIANCE

7.1 Normal Train

It must be confirmed that the current in the traction supply system does not exceed the limits set by the curve in figure 7 in the case of a reed FDM system installed approximately 1m from the section of railway of interest. If it is found that there are significant lengths of traction strengthening conductors alongside the reed FDM system or the cable is laid close to the railway then the limits in figure 10 shall apply. If the limits are not met, route specific analysis of the installed reed FDM systems is required. A less onerous inducing circuit or sectionalisation condition can be used for the calculation if these can be justified on a route by route basis.

7.2 Faulty train

The train current limits defined in figure 7, figure 10 or less onerous limits defined according to the process defined in section 7.1 shall be used as the threshold levels in the train failure analysis. The exceedance rate for train failures can then be determined.

7.3 Safety Targets

The overall exceedance rate shall be found by the failure rate determined in section 7.2. This exceedance rate shall be shown to be ALARP in the context of the safety requirements of the project.

APPENDIX A: Parameters

Parameter:	FDM Cable impedance
Value:	30 Ω /km and 0.3 mH/km
Assumed variation:	This is assumed to be invariant.
Source:	Taken from “Fleet Operation on the 25 kV AC Railway between Crewe and Doncaster – Reed FDM”, ref. I20-dr0458, version b0, dated November 1999.

Parameter:	Receiver transverse voltage limits.
Value:	22.4 mV
Expected variation:	This is assumed to be invariant
Source:	This value is taken as 20% of the drop away value quoted in ““Type RR Reed Frequency Multiplex System Handbook IB138”, GEC General Signal Ltd, dated October 1980.

Parameter:	Capacitance to earth of cable core:
Value:	4 nF/km
Expected variation:	This is assumed to be invariant.
Source:	Taken from “Fleet Operation on the 25 kV AC Railway between Crewe and Doncaster – Reed FDM”, ref. I20-dr0458, version b0, dated November 1999.

Parameter:	Receiver input resistance.
Value:	1.1 k Ω
Expected variation:	This value is assumed to be invariant.
Source:	“Type RR Reed Frequency Division Multiplex System Manual IB165”, GEC General Signal Ltd, dated November 1987.

Parameter:	Transmitter output resistance.
Value:	20 Ω
Expected variation:	This value is assumed to be invariant.
Source:	“Type RR Reed Frequency Multiplex System Handbook IBI 38”, GEC General Signal Ltd, dated October 1980.

Parameter:	Minimum acceptable cable to earth resistance.
Value:	500 k Ω
Expected variation:	This value is assumed to be invariant.
Source:	GK/RT023I Signalling Maintenance Testing, issue 3, dated December 1999.

Parameter:	Maximum gain of line amplifier.
Value:	2
Expected variation:	This value is assumed to be invariant.
Source:	“Type RR Reed Frequency Division Multiplex System Manual IBI 65”, GEC General Signal Ltd, dated November 1987.

APPENDIX B: Assumptions

Configuration

A solid earth fault is present on the reed FDM cable coincident with the minimum acceptable core to earth resistance at the other end.

The infrastructure assumed to be in the worst case condition for inductive coupling.

Supporting References

“Fleet Operation on the 25 kV AC Railway between Crewe and Doncaster – Reed FDM”, ref. I20-dr0458, version b0, dated November 1999.

“Signalling Maintenance Testing”, Railtrack, ref. GK/RT0231, issue 2, dated Feb. 1997.

“Type RR Reed Frequency Multiplex System Handbook IB 138”, GEC General Signal Ltd, dated October 1980.

“Type RR Reed Frequency Division Multiplex System Manual IB 165”, GEC General Signal Ltd, dated November 1987.

Infrastructure failure modes

The reed FDM cable has a solid earth fault at one end and minimum core to earth resistance at the other.

The inducing circuit is based on effectively faulty infrastructure for the length of the exposure.

APPENDIX C: Hazards

Interference current in traction return current

This hazard is covered by the assessment as presented.

Direct induction into track mounted or trackside components

This is the method by which traction current affects the operation of FDM systems.

Particular transient events

No single type or source of transient has been identified as posing a particular risk to reed FDM systems. All transients may contain reed frequency interference therefore train testing must include all foreseeable transient conditions.

Wheelset impedance

The maximum wheelset impedance is not quoted in this document

Vehicle geometry (overhang, wheelbase etc..)

This hazard is not covered by this document.

RSF due to excessive off channel current

This hazard is not covered by this document.

APPENDIX D: Mitigating factors

Pessimistic assumptions

A solid earth fault is present on the reed FDM cable coincident with the minimum acceptable core to earth resistance at the other end.

The infrastructure assumed to be in the worst case condition for inductive coupling.

Assumptions/variations covered by safety margin

Normal variation of reed receiver performance.

Background/coincident sources of interference covered by safety margin

Other rolling stock.

Appendix E: Sources of Technical Work

Sources

The technical work contained in this document is taken from the Class 92 EMC safety case, produced by AEA Technology Rail, ISAs were WS Atkins Rail (secretariat reviewer) and Mott MacDonald.

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019

Issue 1

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Codes of Practice is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

David Wilkinson
Technical & Safety Advisor